

One Predictor/Explanatory Variable, Two Levels

As mentioned at the end of Chapter 3, we'll now move on to cases in which we have a single explanatory variable and a single response variable. In this section, we'll cover the case where the explanatory variable is categorical with only two levels, and the response variable is either categorical or numeric. This means that in this chapter, we'll be focusing on comparing two groups.

Data like these may show up in a spreadsheet like

Categorical Response, Two Levels

First, we'll consider situations in which two categorical variables are measured on each unit in the sample, and each variable has two possible values. In cases like these, typically one variable is considered the response and one variable is considered explanatory. The explanatory variable may be randomly assigned (like whether a subject was assigned to a treatment or control) or it may be merely observed (like smoking status).

The two possible values of the explanatory variable lead to two groups, and we're interested in comparing the population proportions that arise from these two groups. We'll focus on the function of parameters $p_1 - p_2$. The natural estimate of this is $\hat{p}_1 - \hat{p}_2$: the difference in the sample proportions. We'll be constructing hypothesis tests to compare p_1 to p_2 and finding confidence intervals to estimate $p_1 - p_2$. To demonstrate these methods, we'll use an example.

Example: Biologists studying crows will capture a crow, tag it, and release it. These crows seem to remember the scientists who caught them and will scold them later. A study to examine this effect had several scientists wear a caveman mask while they trapped and tagged 7 crows. A control group did not tag any crows and wore a different mask. The two masks did not elicit different reactions from the crows before tagging. Volunteers then strolled around town wearing one or the other of the two masks. The crows scolded a person wearing a caveman mask in 158 out of 444 encounters with crows, whereas crows scolded a person in a neutral mask in 109 out of 922 encounters. Suppose we want to find a confidence interval for the

difference in proportion of crow scoldings between volunteers wearing the caveman mask and those wearing the neutral mask.

For a single proportion, we needed two conditions to be met to ensure the sampling distribution of $\hat{p}$ is approximately normal:

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If these conditions are met, then

We must meet similar conditions to ensure the sampling distribution of $\hat{p}_1 - \hat{p}_2$ is approximately normal:

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If these conditions are met, then

Like before we don't know p_1 and p_2 , so we'll use our best guess. And, like before, our best guess will change depending on whether we're constructing a confidence interval or carrying out a hypothesis test.

How is this going to play out in a confidence interval?

Let's go back to the crows.

How is this going to play out in a hypothesis test?

Again, let's go back to the crows.

We can also do this in R or SAS, but either program will use a different (but also not really) approach. We'll start with R.

```
prop.test(x=c(158,109), n=c(444,922))
```

2-sample test for equality of proportions with continuity correction

```
data:  c(158, 109) out of c(444, 922)
X-squared = 106.11, df = 1, p-value < 2.2e-16
alternative hypothesis: two.sided
95 percent confidence interval:
 0.1867976 0.2884716
sample estimates:
   prop 1    prop 2 
0.3558559 0.1182213
```

From this output, what looks familiar?

What doesn't look familiar?

But is this what we actually tested?

```
prop.test(x=c(158,109), n=c(444,922),alternative="greater",correct=FALSE)
```

2-sample test for equality of proportions without continuity correction

```
data:  c(158, 109) out of c(444, 922)
X-squared = 107.62, df = 1, p-value < 2.2e-16
alternative hypothesis: greater
95 percent confidence interval:
 0.196371 1.000000
```

```
sample estimates:
  prop 1    prop 2 
0.3558559 0.1182213
```

What do you notice?